EEE 491 Lab Assignment
Lab-PCB: Microphone Amplifier and Filter

1. Technical Specifications

An amplifier for a microphone and an anti-aliasing filter circuit and a printed circuit board (PCB) layout of the circuit shall be designed. The amplifier and filter shall use operational amplifiers (OPAMPs) as active components. The microphone, the amplifier, and the filter shall be on a single PCB with maximum area of 25 cm^2.

The design shall use a single supply (no negative voltage). Adjust the gain of the amplifier so that the Lab-PCB output shall interface with the Lab-ADC. The anti-aliasing filter shall use cascaded Sallen-Key low-pass filter (LPF) cascaded stages. The filter shall have 80dB/decade attenuation and its gain from 100 Hz to 3 KHz shall be in -1 to 1 dB range. The gain shall be adjustable by a potentiometer. The 3 dB corner frequency of the LPF shall be in 4 to 4.5 KHz range.

2. Demonstration (Check list)

- Show that your circuit design is implemented on a PCB.
- Present the simulation results of the performance of your design; gain of the amplifier, the filter output voltage level, output offset voltage level, frequency response, and power dissipation. Verify that the technical specifications given above are fulfilled.
- Measure the peak-to-peak voltage levels at the microphone output. Then bypass the microphone connection by removing a series component and connect a signal generator to microphone input. Generate sinusoid signals at various frequencies 50Hz, 100Hz, 300Hz, 1KHz, 2KHz, 4KHz, 10KHz, 20KHz and at various peak-to-peak voltage levels that are close to the previously measured microphone output levels. Observe the output waveforms and verify the technical specifications listed above.
- Speak to the microphone and use your cell phone to generate sinusoid audio signals at the microphone. Verify that output signal is able to swing up to 3.3V peak-to-peak at the output of PCB. Adjust circuit gain if needed.
- Integrate Lab-PCB with Lab-ADC to sample the signals and then run Lab-DEBUG to transfer data to MATLAB. On the MATLAB, plot the samples and measure the signal characteristics both in time and frequency domains.

3. Guidance

First, check the availability of components at the lab before starting to your design.

Use a tool for adjustment of the anti-aliasing filter parameters.

Use a circuit simulator (e.g. LTSpice) and simulate your design and verify your design for the transient response (TRANSIENT analysis) and for the frequency response (AC analysis). Report the results in the final report document.

The Lab-PCB output signal will be interfaced with the A/D converter with 3.3V reference voltage. For AC coupling capacitors use non-polar (non-electrolytic) capacitors. Your microphone (electret type) must be biased together with a filter on its power supply.

Use a schematic and layout editor (e.g. KiCAD). While drawing the layout of the PCB, take precautions for noise. Place the bypass capacitors adjacent to supply inputs of the OPAMPs. Solder the microphone on the PCB instead of connection by cables in order to reduce noise coupling. Try to provide a GND plane wherever possible on both sides of the PCB. Keep the wire lengths between the PCB output and the ADC board input as short as possible. Power supply input has to be filtered by a capacitor.